

Endowus: Transforming wealth management with Google Cloud

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ABOUT ENDOWUS

Endowus uses Google Cloud Kubernetes Engine for the deployment platform, thereby ensuring a more seamless experience for its customers.

Endowus

About Endowus

Launched publicly in April 2019, this Singapore-based investment platform is founded on the idea that investing shouldn't be a costly affair. Endowus believes in using technology to help its customers achieve their financial goals at a much lower cost compared to the traditional way of investing.

Industries: Financial Services &
Insurance

Google Cloud results

- Creates a seamless user experience for customers on the Endowus platform
- Helps Endowus scale to manage tens of millions of dollars for hundreds of clients, just weeks after launch
- Enables Endowus to provide access to better financial products and advice at 30% of the typical industry cost

Location: Singapore

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lower
inves
costs

Google Cloud (<https://cloud.google.com/>)

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Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

- Helps
Endowus
make more
accurate
marketing
decisions
with
BigQuery
- Allows
integration of
data from
different
platforms
hosted by
other
organizations

For many, investing feels like a tedious and costly process, but Singapore-based investment platform Endowus (<https://endowus.com/>) aims to simplify the process and even make it more enjoyable. The company offers a one-stop shop where investors can get advice and a range of investment options without paying a high fee. "With an experienced team in the investment management industry, we want to become a platform that provides access to the best products and advice at the lowest cost possible," says Gregory Van, co-CEO and COO of Endowus.

Pushing the boundaries of technology to benefit its customers by keeping costs low, Endowus' strategy is to remain an asset-light platform by keeping a lean team that understands how to use technology

to scale the business. "The cost of investing has traditionally been very high due to the manual processes involved. We believe most of this can be automated with the help of technology," he adds.

Laying the right foundations with Google Kubernetes Engine

"We were looking for the quick and cost-efficient solution to build our microservices platform. After careful evaluation, we went with Google Kubernetes Engine as it offers a production-ready cluster with all the necessary

tooling and support.”

—Joo Won Lee, CTO,
Endowus

With clients that range from large institutions to individuals, Endowus partners with large financial institutions and fund managers to provide a wide variety of portfolio offerings to its customers. As an investment platform, it's crucial for Endowus to build and deploy new features quickly while maintaining good visibility of its internal processes. As such, it turned to Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>) (GKE).

Since deploying GKE, the company has been able to run quicker deployments and feedback loops. "Initially, the whole platform was basically deployed in one environment. We used different open source backend technologies on the platform that provided no visibility on what was going on in the system. Soon after, we decided to rebuild the existing system using a microservices architecture. Deploying it on Google Kubernetes Engine made the most sense," says CY Koh, Dev Ops Lead at Endowus.

"We were looking for the quickest and most cost-efficient solution to build our microservices platform on top of Kubernetes," adds Joo Won Lee, CTO at Endowus. "We looked at other cloud service providers with hosted Kubernetes solutions and also considered on-prem options. After careful

evaluation, we went with Google Kubernetes Engine as it offers a production-ready cluster with all the necessary tooling and support to make sure packages and versions are always up-to-date."

"With clear processes and basic templates set up, Google Kubernetes Engine makes it very easy for developers to understand what is going on under the hood. This removes uncertainties about the deployment process, and developers can easily deploy new changes and services."

—CY Koh, Dev Ops Lead,
Endowus

Endowus now has more than 40 microservices with hundreds of pods operating in each environment.

"With clear processes and basic templates set up, Google Kubernetes Engine makes it very easy for developers to understand what's going on under the hood. This removes uncertainties about the deployment process, and developers can easily deploy new changes and services," says CY. As a result, developers are now able to focus on building and pushing out new features with confidence that the system will work as expected.

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*—Gregory Van, co-CEO
and COO, Endowus*

As an investment platform, the company is constantly feeding live market data into its platform and providing real-time updates to its customers on how their investments are performing. To do so, its platform interacts with various systems in the finance world, including legacy systems as far back as fax machines. "Different institutions have different systems and requirements. It's our job to translate all of that into a seamless experience for the end customer through our platform. Google Kubernetes Engine gives us the ability to improve that experience and respond to customer feedback at a faster rate," says Gregory.

Soon, the company will launch its newest and largest feature to date, the Central Provident Fund (CPF) investment portfolio. Like its other offerings, this feature depends on GKE to ensure a smooth and easy process. "With a horizontally scalable microservices architecture, the platform is easy to scale up to meet the demands of our customers," says CY.



Charting the future with BigQuery

Endowus plans to run its first marketing campaign soon, and will be using BigQuery

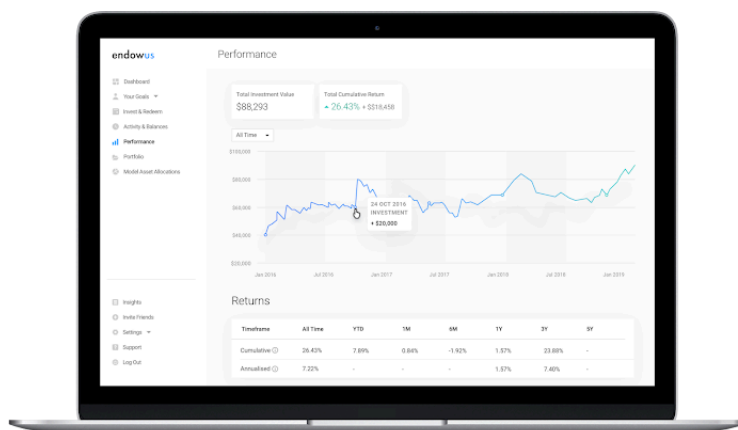
(<https://cloud.google.com/bigquery/>) to provide the analytics that will drive it. "The first stage will be to figure out customer behavior, their demographic profiles, how they invest, and how they respond to market changes. Then, we will look at our existing customers to learn more about their interests and needs, and finally see how we can measure our marketing efforts," says CY. He adds that the data from BigQuery will be crucial to its decision-making process in terms of resource allocation, as it will

help ensure that marketing budgets are efficiently spent.

Meeting security standards in the finance world

As a Monetary Authority of Singapore (MAS) licensed investment platform, Endowus is required to meet certain security standards. "By virtue of being on Google Cloud, we are compliant with a lot of regulatory requirements without having to build things from scratch," says Gregory.

From access controls and encryption to network perimeter defence, the transparent structure of GKE has made it easy for engineers to address the specific rules that are applied to certain servers.



Braving the future with cutting-edge technology

With [Google Cloud](https://cloud.google.com/) (https://cloud.google.com/) as a partner, Endowus is now able to constantly develop

new products with a very quick time to market. "With the capabilities of Google Cloud and our highly skilled and passionate team members, our 10 developers are doing the work that would typically take 30 developers," says Gregory.

He adds that having a partner like Google to support the team is crucial in order for the company to move forward successfully. "The Google sales and customer engineers are extremely knowledgeable about their product, and their supportive nature really makes Google a great partner to work with."

Endowus already has plans to expand beyond the shores, but will first focus on the Singapore market and its latest CPF investment portfolio offering. "We have a lot of products to offer the Singapore community, and we are glad to be able to do this with Google Cloud. Importantly, it speaks our language when it comes to understanding which products can best meet the needs of our company, and that fills us with confidence," says Gregory.

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our team to
see how
Google Cloud
can help your

business.